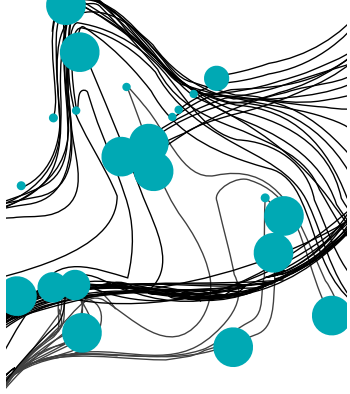


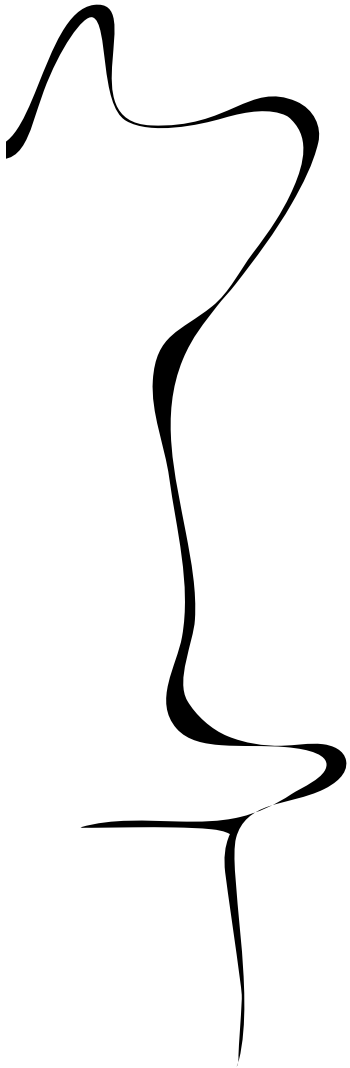


Modelling & Simulation

Final Assignment

Name	Student-number
Luce Zoontjens	s2678934
Jacco Bluemink	s3309398
Kristian Plat	s3753794
Steven Pennock	s3759016

Date:
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UNIVERSITY OF TWENTE.

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1. Introduction

This report documents the modelling, actuation, and control of a 3 DoF robot arm which has a cascaded control structure with impedance, torque, and current control.

Instead of designing an arbitrary robot arm, we chose to base our model on an existing one: the Franka Research 3 (FR3) [1]. It has a well-documented GitHub [2] with most of the needed robot parameters (e.g. link dimensions, masses, inertias, see Table 5) which makes our model more realistic. The FR3 has 7 joints, but we reduced this to a yaw-pitch-pitch chain by locking joints 3 and 5 at 0 rad , and removing joints 6 and 7. This preserves the 3D reachability of the end-effector, while keeping to the required 3 DoF.

The actuation and control architecture follows the cascaded structure, which was learned throughout the course. Each joint is driven by a brushless DC motor and gearbox in series with a compliant element (the Series Elastic Actuator (SEA)). The SEA serves two purposes: it acts as a torque sensor by measuring spring deflection, and it filters out torque spikes before they reach the gearbox. The joints are controlled by 3 nested loops: the inner current control loop, followed by the torque control loop, and the outer Cartesian impedance control loop. The torque loop uses the SEA spring deflection as a torque sensor. The impedance loop computes the desired joint torques from the end-effector position and the velocity error.

Because Franka does not publish the specifics of their used motor and gearbox, we choose a pair that matches the robot's published torque and speed envelope. This motor-gearbox pair is the Maxon EC Frameless DT 50M motor [3] and Harmonic Drive CSF-20-120-2UH gearbox [4].

Several modelling choices shaped the rest of the model. First, we used 20-sim's 3D Mechanics Editor [5] for the rigid-body dynamics of the arm instead of deriving the Euler-Lagrange equations. However, an independent kinematic model for the forward kinematics is used in the Cartesian impedance controller, which we derived analytically. This way, the controller relies on a closed-form Coriolis matrix and a gravity feedforward term instead of black-box estimates. The gravity feedforward ensures the impedance spring does not need to support the arm's own weight, by keeping the impedance gains low for a safe and compliant interaction. A second choice that was made was not to use the off-diagonal inertia terms published by Franka. The 3D Mechanics Editor only supports diagonal inertia tensors, and since the off-diagonal terms are pretty small relative to the diagonal ones, this is a reasonable simplification. It does mean the rigid-body dynamics are slightly approximated, which is acknowledged as a modelling assumption. Third, all 3 the control loops are implemented in discrete time.

The cascaded discrete time structure was built up incrementally to avoid oscillations and integration errors since the loops are prone to that. Starting from a single actuator, the inner current loop was tuned first at 10 kHz using 20-sim's built-in parameter optimisation with an ITAE (Integral Time Absolute Error) cost function to determine the optimal values. We used this cost function because it not only ensures the target is set quickly, but also ensures there is little to no overshoot and that small oscillations after settling are taken into account. The torque and impedance loops were then added in sequence. During torque controller tuning, a persistent steady-state offset was found, and after investigating, it was found that the cause is the Coulomb friction in the drivetrain model. To compensate for this a friction observer [6] was implemented that estimates and compensates the Coulomb friction for this.

With this model, several simulation experiments were done. For verification, the subsystems were isolated to check their basic behaviour before integrating them together. For example, the unactuated arm should swing under gravity as a pendulum would. For validation, it was checked if the full model could perform a repeated reaching task. This task consists of 4 Cartesian target points, which are generated through a minimum-jerk interpolation and then converted back to the Cartesian reference trajectory. Next, a self-made external force is applied at the end-effector to demonstrate that the arm behaves as a proper spring-damper and gives way under contact. A video of this experiment, with the arm held at a 0.5 rad pose under external loading, is available at <https://youtu.be/wPj-Wpw-qU>.

2. Model overview

In Figure 1 the model is captured in a broad overview. Setpoints are shaped into a minimum jerk trajectory (MJT) and sampled with AD-converters. The impedance controller turn the end-effector (EE) position and velocity error into desired joint torques, which drives the joint actuators. Each actuator contains torque and current loops, the electrical and rotor dynamics, gearbox and the series electric actuator (SEA) spring, and transmits its torque through to the 3D robotarm. The resulting joint positions are fed back to the controller, which closes the loop. The detailed signals and feedback paths are discussed in the following subsections.

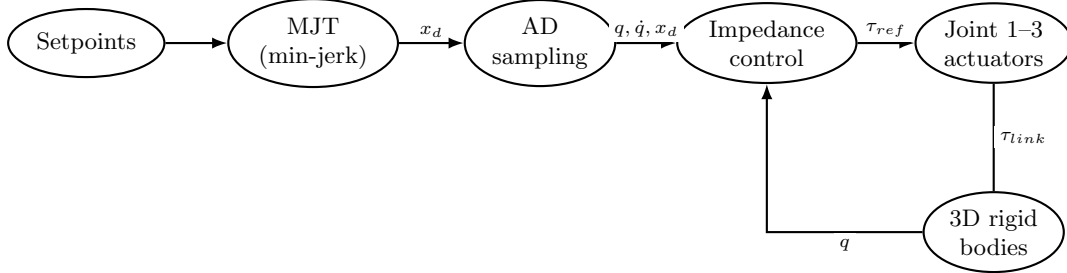


Figure 1: Word bond graph overview of the full model. Single arrows are signals, the double line is a mechanical power bond. See `G01_finalmodel.emx`.

2.1 Robot Arm Geometry and Material Properties

The 3 DoF configuration was chosen as a yaw-pitch-pitch chain, so the first joint rotates about the vertical axis and the two remaining joints both rotate about horizontal axes in the same plane. This was selected because it gives full 3D reachability of the end-effector with the minimum number of joints. The yaw joint sweeps the arm across the horizontal plane, while the two pitch joints control the reach and height within that plane. A pitch-pitch-pitch chain would lack yaw reachability entirely, and a yaw-yaw-pitch or something similar would introduce out-of-plane kinematics that could complicate the inverse kinematics without adding any useful workspace [7]. The yaw-pitch-pitch structure also maps naturally onto the FR3's first three active joints, which makes the reduction from 7 to 3 DoF more straightforward. Instead of designing an arbitrary arm and deriving the geometry and inertia based on material and link volume, our choice was to reverse-engineer an existing robot: the Franka Research 3 (FR3) [1]. The FR3 is a 7-DoF manipulator used in research whose complete kinematics and inertial description is published in the official `frank.description` package [2]. It is *assumed* that the values in the official package are an accurate representation of the physical robot and are treated as ground truth of the designed model. This choice keeps the model realistic without the need of estimating material properties, and shifts the focus to the modelling effort on the actuation and control architecture.

Since the desired model only has 3 DoF instead of the 7 of the FR3, the kinematic chain had to be reduced. The yaw-pitch-pitch chain is obtained by keeping FR3 joints 1, 2 and 4 active and locking joints 3 and 5 at 0 rad . Locking these joints means the links on either side of them no longer move relative to each other, so they can be treated as a single rigid body. FR3 links 2 and 3 fused into body 2, and links 4 and 5 into body 3. The masses, centres of mass and inertia tensors of the merged bodies are not summed, but combined about the new common centre of mass by using the parallel axis theorem. This preserves the rigid-body behaviour. All the parameters are directly taken from the FR3 inertial data listed in Table 5.

Because of the reverse-engineering, the link lengths are not free design parameters either. They follow directly from the joint frame offsets in the FR3 description format (Table 5). Joint 1 is offset 0.333 m along the z -axis from the base, and body 3 carries the combined offset of the merged forearm, giving the arm its characteristic double-kink where the elbow steps out 0.082 m in x and the forearm returns to the vertical axis. Verified at the zero configuration, the joint frames sit at $[0, 0, 0.333]$, the elbow at $[0.0825, 0, 0.649]$ and the end-effector at $[0, 0, 1.033]$, which matches the nominal FR3 geometry.

The geometric and inertial properties are not implemented as hand-derived Euler-Lagrange equations, but

mapped into the 20-sim 3D Mechanics Editor [5]. Each of the three bodies is given its mass, centre-of-mass offset and inertia tensor from Table 5, and the joints are placed at the corresponding frame offsets with the correct rotation axes (z for joint 1, y for joints 2 and 3). The editor then generates the rigid-body dynamics automatically, which avoids manual derivation errors and provides the multi-body plant that the control loops act on.

The 20-Sim 3D Mechanics Editor was used for the modelling of the multi dynamics robot arm. The model is made with a base and 3 independent moving links. The freedom of movement between these links is made with joints; they link 2 bodies together and have one degree of freedom. The joints in the 3D model have the following implementation:

- **Joint 1 (Yaw):** Connects the base with the first link and rotates around the Z-axis
- **Joint 2 (Pitch):** Connects the first link with the second and rotates around the Y-axis.
- **Joint 3 (Pitch):** Connects the second link with the third and rotates around the Y-axis.

The dynamics properties (mass, inertia) of the Franka Robot are assigned to each link. To simulate the effect of gravity on the robot arm, a global gravity vector is configured with $-9.81m/s^2$ along the Z-axis.

To integrate the mechanical model with the external controls, the 3D model is equipped with input and output ports. The model uses the input torque to control its motors and outputs the motor position to the control loops.

In Figure 2 the 3D model is shown in starting position. Just like the real robot arm on which this model is based on, the joints have a orientation offset.

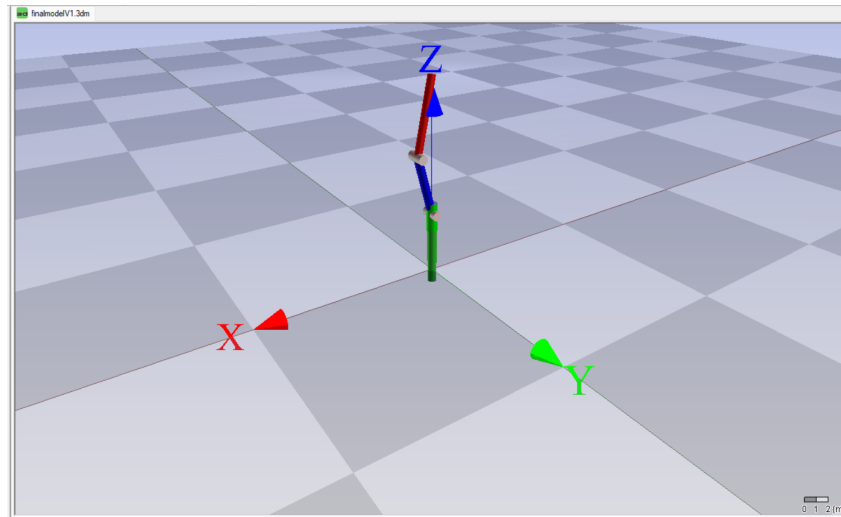


Figure 2: 3D model in the 20-sim 3D Mechanics Editor.

2.2 Actuator Model

To build an accurate simulation model of the robotic joint drivetrains, the specifications of the Franka Research 3 (FR3) were analysed. The official GitHub page [2] and data sheet [8] define limits like peak torques and maximum velocities[9][10]. Unfortunately the manufacturer keeps the details hidden, leaving out the exact make, model, or technical data sheets of the embedded motors and gearboxes.

Each joint is driven by a brushless DC motor connected in series with a harmonic drive gearbox and a SEA.

2.2.1 Motor

A brushless DC motor converts electrical power into mechanical power through electromagnetic transduction. The motor constant K_t describes both sides of this conversion: the back-EMF voltage which is

induced by the spinning rotor, and the torque produced by the current carrying winding:

$$V_{\text{emf}} = K_t \omega_m, \quad \tau_m = K_t i$$

The electrical side of the motor is described by Kirchhoff's voltage law on the armature circuit. The winding has resistance R_w and inductance L_w , and the back-EMF acts as a voltage source opposing the supply voltage:

$$V = R_w i + L_w \frac{di}{dt} + K_t \omega_m$$

On the mechanical side, Newton's second law is applied on the rotor and gives:

$$J_m \dot{\omega}_m = \tau_m - \tau_{f,m} - \tau_{\text{gearbox}}$$

where J_m is the rotor inertia, $\tau_{f,m}$ is the rotor friction torque and τ_{gearbox} is the reaction torque from the gearbox.

2.2.2 Gearbox

The harmonic drive gearbox is a speed reducer with ratio N . It scales the torque up and the angular velocity down between the motor shaft and the joint output:

$$\tau_{\text{out}} = N \tau_{\text{in}}, \quad \omega_{\text{out}} = \frac{\omega_{\text{in}}}{N}$$

The gearbox has its own rotational inertia J_g and internal friction $\tau_{f,g}$ which both refer to the motor side of the transmission.

2.2.3 Series Elastic Actuator (SEA)

A SEA places a finite-stiffness spring in between the gearbox output and the load instead of connecting them rigidly [11]. This design choice gives 2 benefits to our model: The first is torque sensing. Because the spring obeys Hooke's Law, the amount of torque that passes through can be measured by the deflection of the spring: $\tau = K_{SEA} \Delta\theta$ ($\Delta\theta = \theta_{\text{gear}} - \theta_{\text{link}}$). This turns the difficult problem of torque measurement into the easier problem of angle measurements.

The second benefit is mechanical filtering. The spring also acts as a buffer between the fast spinning motor and rotor on one side and the much slower moving load/link on the other. It smooths out the torque spikes before they reach the gearbox [11].

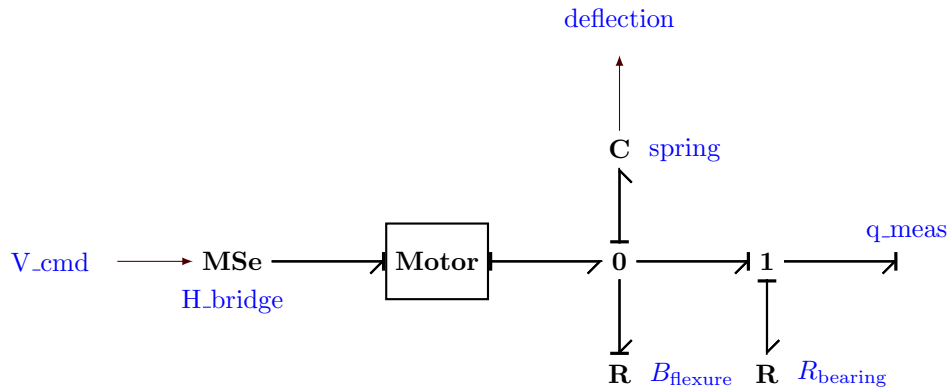
The choice of spring stiffness is a trade-off [12]. A softer spring deflects more for a given torque, which makes it more sensitive. This is better for force and shock absorption, but it also limits how fast the torque loop can be controlled. On the opposite, a stiffer spring is faster but less sensitive. To measure the deflection we chose to use a strain gauge sensor on a flexure in the SEA. This provides a high-resolution readout, which makes it possible to use a higher K value for the SEA spring. We chose $K_{SEA} = 13000 \text{ Nm/rad}$ as a middle ground. It is stiff enough that the torque loop stays comfortably faster than the outer Cartesian loop, but still soft enough so it produces a measurable deflection under the torques the arm generates.

The internal spring torque τ_s is governed by:

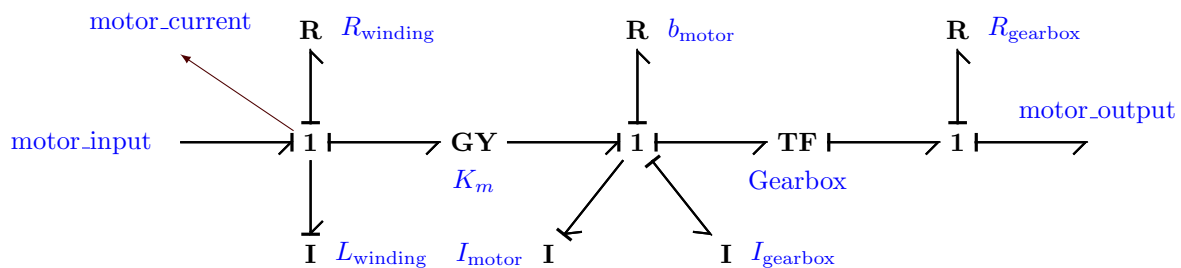
$$\tau_s = K_{\text{sea}}(\theta_{\text{gear}} - \theta_{\text{link}}) \quad (1)$$

where K_{sea} is the rotary spring stiffness, θ_{gear} is the output angle of the gearbox, and θ_{link} is the actual joint position.

Two other dissipative elements act on the joint output side of the drivetrain. The output bearing contributes viscous friction, which value is taken from the Franka robot. The compliant element of the SEA has a flexure damper in parallel with the spring. Without this term the SEA spring and the 2 inertias on either side of it would form a lightly damped second-order resonator. This flexure damper provides the dissipation that prevents this oscillation. Its value was chosen to give a well damper torque loop response at the chosen spring stiffness.



Bondgraph of the SEA.



Bondgraph of motor model in the SEA.

2.2.4 Friction Model

Two physical models contribute to friction at every rotating shaft in the drivetrain. Viscous friction produces a torque proportional to velocity, which represents the shear losses and it dominant at higher speeds. Coulomb friction produces a constant torque that opposes the motion regardless of the speed. This represents the contact between surfaces and is dominant near zero velocity. Both terms are needed since viscous alone can not model the velocity independent stiction that resists motion at near zero speed, while Coulomb alone can not capture the speed dependent losses at higher speeds. The combined friction torque at a shaft rotating at angular velocity is therefore:

$$\tau_f(\omega) = b\omega + T_c \tanh(\gamma\omega)$$

The ideal Coulomb term $T_c \text{sgn}(\omega)$ has a discontinuity at $\omega = 0$, which produces some numerical issues. Therefore the hyperbolic tangent $\tanh(\gamma\omega)$ is used which approximates this smoothly. With $\gamma = 5000$ the transition is almost instantaneous on the timescales of the simulation while it still remains numerically tractable near zero velocity.

Motor friction parameters The friction model has 2 free parameters: b_m and $T_{c,m}$, but the Maxon datasheet for the EC frameless DT50M only provides one usable constraint: the peak electrical-to-mechanical peak of 92%. To uniquely determine both parameters from this one number, a second condition is needed. The chosen condition is a 50/50 power split; at the speed of peak efficiency ω^* , viscous and Coulomb friction dissipate equal power:

$$P_{\text{viscous}}(\omega^*) = P_{\text{Coulomb}}(\omega^*) \implies b_m \omega^{*2} = T_{c,m} \omega^* \implies T_{c,m} = b_m \omega^*$$

This is a neutral assumption: because of the absence of data (which would possibly favour one over the other) we assume they contribute equally at the operating point. It does not mean the 2 parameters value are numerically equal but that their power contributions are. With this constraint, we can solve for b_m and $T_{c,m}$: at peak efficiency, the rate of copper lose equals the rate of friction loss:

$$i^{*2} R_w = \tau_f(\omega^*) \omega^*, \quad i^* = \frac{V - K_t \omega^*}{R_w}$$

Substituting $\tau_f = b_m \omega + T_{c,m} \tanh(\gamma\omega)$ and the constraint $T_{c,m} = b_m \omega^*$ into the equation above gives a self consistent system. ω^* is the speed at which efficiency is maximised, and at that same speed must satisfy the 50/50 power contribution. Solving numerically then gives:

$$b_m = 1.258 \times 10^{-5} \text{ Nms/rad}, \quad T_{c,m} = b_m \omega^* = 5.51 \times 10^{-3} \text{ Nm}, \quad \omega^* = 438 \text{ rad/s}$$

Figure 3 illustrates the Coulomb-viscous ratio. It shows that the viscous and Coulomb power curves cross at exactly $\omega^* = 438 \text{ rad/s}$, which coincides with the 92% efficiency.

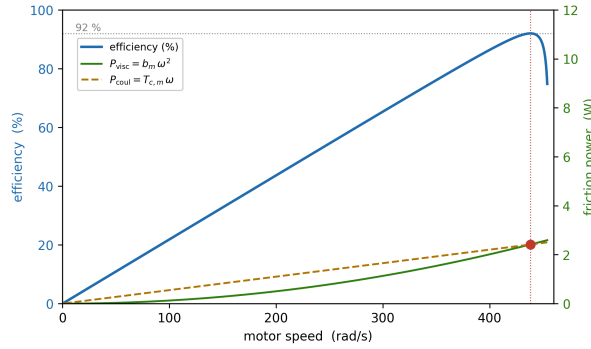


Figure 3: Motor friction parameter derivation

Gearbox friction parameters The same friction model of the motor is applied to the gearbox. The parameters are determined differently. The Coulomb friction value is taken directly from the harmonic drive datasheet and the viscous friction value is tuned so that the simulated transmission efficiency matches the datasheet value of 80%.

2.2.5 Component Selection

To work around this limitation, a component-matching approach was used based on the available performance data. Cross-referencing the required continuous torques and maximum speeds with commercially available parts led to the selection of an alternative. The available parameters can be seen in Table 1.

Table 1: Available Actuator and Dynamics Parameters

Parameter	Joint 1 & Joint 2 (Base & Shoulder)	Joint 3 (Elbow)	Unit
Active Original Joint	Joint 1 (Yaw) & Joint 2 (Pitch)	Joint 4 (Pitch)	—
Gearbox Ratio	120.0	120.0	—
Motor Inertia	4.206399×10^{-5}	3.211626×10^{-5}	$\text{kg} \cdot \text{m}^2$
Output Torque (Continuos)	87.0	87.0	$\text{N} \cdot \text{m}$
Output Speed (limit)	2.62	2.62	rad/s

For the actuation, all three joints were outfitted with identical Maxon motors. On the real FR3, the elbow motor (Joint 4) is actually slightly different from the larger Joints 1 and 2 motors. However, because the performance differences between them are small, the exact same Maxon motor was chosen for all three joints to keep the simulation design uniform and easier to analyze.

To achieve the correct speed-to-torque conversion and meet the required 1:120 reduction ratio, these motors were paired with a uniform set of gearboxes. Using identical gearboxes directly matches the FR3, which uses the same gearbox type for these joints. The entire 3-DOF drivetrain remains completely uniform, reducing modeling complexity while still mirroring the capabilities of the original robot.

The *Maxon EC frameless DT 50 M* [3] brushless DC motor was chosen because its specifications match the FR3 data. By comparing the required torque limits, maximum speeds, and motor inertia with available commercial parts, this specific motor proved to be a good fit. It delivers the high torque and

low inertia needed to accurately simulate the quick accelerations of the real robot. Using the exact same motor for all three active joints keeps the simulation simple and uniform.

The *Harmonic Drive CSF-20-120-2UH* [4] gearbox was chosen because it matches the original mechanics of the FR3 robot. The baseline data requires a reduction ratio of 1:120 for all active joints. Using a Harmonic Drive system is important because the real Franka robot also relies on these gearboxes. These gearboxes are the industry standard for high-end robots because they have zero backlash (no mechanical play), high torque capability, and a very compact, lightweight design [4].

To ensure that the 20-sim model mimics a real system, energy losses and efficiency constraints are included. Instead of assuming ideal components, the internal friction parameters in the model were calibrated using the official manufacturer catalogs for the selected hardware.

According to the official Maxon documentation [3], the *EC frameless DT 50 M* motor operates at a peak electrical-to-mechanical efficiency of 92%. Our estimation is that a 50/50 parameter split was used to tune the motor's internal viscous friction (b_m) and Coulomb friction (τ_{cm}). A more in-depth analysis is done in Appendix A.

For the transmission, the official Harmonic Drive catalog [13] indicates that the *CSF-20-120-2UH* series operates at an efficiency of approximately 80% under nominal speeds and loading conditions. To replicate this behavior in the 20-sim submodel, the gearbox Coulomb friction (τ_{cg}) was taken directly from the datasheet, while the gearbox viscous friction (b_g) was tuned to hit the 80% operating efficiency target. A more in-depth analysis is done in Appendix B.

Incorporating these specific catalog profiles guarantees that the simulated 3-DOF robot accurately accounts for realistic torque drops, dampening, and power dissipation based on the chosen hardware specifications. For an overview of all selected parameters see Table 2.

Table 2: Maxon EC frameless DT 50 M motor and Harmonic Drive CSF-20-120-2UH gearbox

Submodel	Symbol	Parameter	20-sim Name	Unit	Joint (1, 2, 3)
<i>Motor</i>	-	Type (Maxon Motor)	-	-	EC frameless DT 50 M
	K_t	Motor Constant	Motor\r	Nm/A	0.105
	R	Armature Resistance	R.winding\r	Ω	0.762
	J_m	Standalone Rotor Inertia	I_motor\i	$\text{kg} \cdot \text{m}^2$	9.88×10^{-6}
	J_w	Stator Winding Inertia	I.winding\i	$\text{kg} \cdot \text{m}^2$	5.71×10^{-4}
	b_m	Motor Viscous Friction ¹	b_motor\b_motor	Nms/rad	1.27×10^{-5}
	τ_{cm}	Motor Coulomb Friction	b_motor\Tc_motor	Nm	0.0057
<i>Gearbox / Dynamics</i>	γ_m	Motor Smoothness	b_motor\gamma	-	5000.0
	-	Type (Harmonic Drive)	-	-	CSF-20-120-2UH
	N	Ratio	Gearbox\r	-	120
	J_g	Gearbox Inertia	I_gearbox\i	$\text{kg} \cdot \text{m}^2$	1.93×10^{-5}
	b_g	Gearbox Viscous Friction ²	R_gearbox\b_gear	Nms/rad	1.75×10^{-4}
	τ_{cg}	Gearbox Coulomb Friction	R_gearbox\Tc_gear	Nm	0.038
	γ_g	Gearbox Smoothness	R_gearbox\gamma	-	5000.0

¹ Tuned using a 50/50 parameter split to achieve a maximum efficiency of 92%, matching the official Maxon datasheet thresholds.

² Coulomb friction was given in the datasheet, so we tuned the viscous friction to match a realistic 80% (as stated in [13] the efficiency is expected to be between 80 and 90%) operating efficiency under nominal joint speed and torque conditions.

2.3 Cascaded Control

Our controller consists of 3 loops nested inside each other, running at two different speeds: a fast inner current loop (10 kHz), and the intermediate torque and outer impedance control (1 kHz).

The reason for this nesting is that it lets each loop almost ignore what the faster loop inside it is doing. As long as the inner loop is fast enough, by the time the outer loop checks in again, the inner loop has already settled wherever it was told to go. So, from the torque loop's point of view, the current loop appears to deliver the requested current instantly, and from the impedance loop's point of view, the torque loop appears to deliver the requested torque instantly [14].

This speed separation was built directly into the sample rates. The current loop runs at 10 kHz, while the torque and impedance loops both run at 1 kHz. This is the same inner/outer split introduced earlier in this course's motor control assignment, where the current loop is described as typically running an order of magnitude faster than the loop around it. The same principle also shows up in real hardware. For example, on the DLR light-weight robot, whose joint-level loop runs at 3 kHz against a 1 kHz Cartesian loop [15].

2.3.1 Cartesian Impedance Control

Impedance control controls the dynamic relation between motion and force at the end-effector. Instead of driving the arm to a setpoint and fighting anything that gets in the way, the controller makes the end-effector behave like a virtual spring and damper around a moving target: $F = -(D\dot{e}_x + Ke_x)$. When you push on it, it pushes back proportionally to displacement and resists proportionally to velocity. This what lets the arm 'give way' safely on contact instead of resisting at full strength. The stiffness and damping gains were chosen as $K = 400$ N/m and $D = 65$ Ns/m. Since gravity is handled separately by a feedforward gravity compensation term g_{ff} , the impedance spring does not need to support the arm's own weight. Therefore, K is chosen based on that it is high enough that the steady-state position error at each setpoint stays within a few millimetres, but also low enough that the arm gives way compliantly under external contact forces instead of resisting them. D is set to give a critically damped response at the chosen stiffness [16]. For a second-order impedance the critical damping condition is $D = 2\sqrt{K \cdot M_{\text{eff}}}$, where $M_{\text{eff}} = (J(q)M(q)^{-1}J(q)^T)^{-1}$ is the effective end-effector mass projected by the Jacobian. At the nominal reaching pose this evaluates to around 2.64 kg, which gives $D = 2\sqrt{400 \times 2.64} \approx 65$ Ns/m, which prevents overshoot when moving to a new setpoint.

To use this, first the end-effector's actual position as a function of the joint angles is needed. This is obtained from the forward kinematics by composing each joint's homogeneous transform from base to end-effector and reading off the position column:

$$R_{z,1} = \begin{pmatrix} \cos q_1 & -\sin q_1 & 0 \\ \sin q_1 & \cos q_1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, R_{y,2} = \begin{pmatrix} \cos q_2 & 0 & \sin q_2 \\ 0 & 1 & 0 \\ -\sin q_2 & 0 & \cos q_2 \end{pmatrix}, R_{y,3} = \begin{pmatrix} \cos q_3 & 0 & \sin q_3 \\ 0 & 1 & 0 \\ -\sin q_3 & 0 & \cos q_3 \end{pmatrix}$$

$$T_i = \begin{pmatrix} R_i & p_i \\ 0^T & 1 \end{pmatrix}, T = T_1 T_2 T_3 T_{ee}$$

$$x = \left[T \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right]_{1:3}$$

The relationship between joint velocities and end-effector velocity is given by the geometric Jacobian, whose columns are the cross product between each joint's rotation axis and the vector from joint to end-effector.

$$J(q) = \begin{pmatrix} \hat{z}_0 \times \mathbf{x} & \hat{z}_1 \times (\mathbf{x} - \mathbf{p}_2) & \hat{z}_2 \times (\mathbf{x} - \mathbf{p}_3) \end{pmatrix}$$

$$\dot{x} = J(q)\dot{q}$$

The virtual force F from the impedance law is not something the motors can directly apply, it has to become joint torques, which is done with help of the Jacobian ($\tau = J^T F$). Of course, more dynamics work on the arm, such as gravity and centrifugal forces. These dynamics are fed forward in the equation:

$$F = -(D\dot{e}_x + Ke_x)$$

$$e_x = x - x_d, \dot{e}_x = \dot{x} - \dot{x}_d$$

$$\tau_d = J^T F + C(q, \dot{q})\dot{q} + g_{ff}$$

To compensate for the gravity that works on the arm, the following gravity compensation is done:

$$g_{ff} = J_1^T \begin{pmatrix} 0 \\ 0 \\ -m_1 g \end{pmatrix} + J_2^T \begin{pmatrix} 0 \\ 0 \\ -m_2 g \end{pmatrix} + J_3^T \begin{pmatrix} 0 \\ 0 \\ -m_3 g \end{pmatrix}$$

Here J_1 , J_2 , J_3 are the Jacobians that map joint velocities to the linear velocity of each link's centre of mass respectively, since that is where gravity acts.

As the arm moves, the changing configuration shifts how mass is distributed, which generates velocity dependent forces: the Coriolis and centrifugal effects. These arise because the inertia matrix $M(q)$ is configuration dependent: accelerating one joint while another is moving produces a reaction torque on the other joints, even without any external contact. If this is left uncompensated, these forces act as disturbances on the impedance controller and cause the end-effector to deviate from its path during faster motions. The feedforward term $C(q, \dot{q})\dot{q}$ cancels them before they can cause a tracking error. The inertia matrix $M(q)$ is derived from the kinetic energy of the three links, and uses the same masses, centres of mass and inertia tensors as listed in Table 5.

The Coriolis matrix $C(q, \dot{q})$ is derived analytically from the inertia matrix $M(q)$ using the standard Christoffel symbol formula [17]:

$$c_{kj} = \sum_i \frac{1}{2} \left(\frac{\partial M_{kj}}{\partial q_i} + \frac{\partial M_{ki}}{\partial q_j} - \frac{\partial M_{ij}}{\partial q_k} \right) \dot{q}_i$$

For the yaw-pitch-pitch chain, the inertia matrix $M(q)$ has only one configuration dependent entry: M_{11} , which is the inertia seen by the yaw joint, which changes as the two pitch joints swing the arm mass in and out. The pitch-pitch coupling terms are constant because the two pitch joints share the same rotation axis. Evaluating the Christoffel symbols for this gives:

$$C(q, \dot{q}) = \begin{pmatrix} \frac{1}{2} \left(\frac{\partial M_{11}}{\partial q_2} + \frac{\partial M_{11}}{\partial q_3} \right) \dot{q}_3 & \frac{1}{2} \frac{\partial M_{11}}{\partial q_2} \dot{q}_2 & \frac{1}{2} \frac{\partial M_{11}}{\partial q_3} \dot{q}_1 \\ -\frac{1}{2} \frac{\partial M_{11}}{\partial q_2} \dot{q}_1 & -m_3 L_2 \frac{L_3}{2} s_3 \dot{q}_3 & -m_3 L_2 \frac{L_3}{2} s_3 (\dot{q}_2 + \dot{q}_3) \\ -\frac{1}{2} \frac{\partial M_{11}}{\partial q_3} \dot{q}_1 & m_3 L_2 \frac{L_3}{2} s_3 \dot{q}_2 & 0 \end{pmatrix}$$

where $s_3 = \sin q_3$. The first row captures the reaction torques on the yaw joint caused by pitch motion which changing the arm's moment of inertia around the vertical axis. The off-diagonal terms in the other rows capture the coupling between the two pitch joints through the elbow mass m_3 , which depends on the elbow angle q_3 via s_3 .

2.3.2 Torque Control

The torque controller does 2 things at once. It feeds forward the desired torque directly, and it adds a small correction based on the torque error. The measured torque is obtained from the SEA spring deflection via Hooke's law: ($\tau_{meas} = K_{SEA} \Delta\theta$). The torque error is then given by:

$$e_\tau = \tau_{ref} - K_{SEA} \Delta\theta$$

PD feedback A discrete PD controller acts on the torque error. PD control was chosen because the SEA in combination with the 2 inertias, gearbox/rotor inertia and link inertia, creates a lightly-damped spring-mass-spring system with has an oscillatory element. Therefore, a derivative term is needed to suppress this resonance. An integral term would remove the steady-state offset, but since the feedforward term already is doing that, there is no need for it. The discrete PD is implemented as:

$$u_D[k] = \frac{1}{T_s + \tau_D \beta} \left(\tau_D \beta u_D[k-1] + \tau_D k_p (e_\tau[k] - e_\tau[k-1]) + T_s k_p e_\tau[k] \right)$$

where $k_p = 10$, $\tau_D = 2 \times 10^{-5}$ s and $\beta = 0.3$ is a constant that filters the derivative to limit noise amplification. The full torque command is then:

$$\tau_{cmd} = \tau_{ref} + u_D + \hat{\tau}_{fric}$$

Friction observer When tuning the torque controller, we noticed a constant offset with the target torque. After some investigating on what could be causing this, we found it to be the Coulomb friction we added to our model. We came to this conclusion because when we set the Coulomb friction to 0, the constant error disappeared. Now this could be solved by adding an I term, making the controller a PID; however, this could cause unwanted behaviour when the error stays high for too long and therefore causes a big overshoot due to the I term. Now this overshoot from the I term is very much not wanted when the robot arm should be handling something carefully. We therefore set out to find a way to compensate for this in a different way. Now, the easiest way we found is to just add the Coulomb friction, which we already know in our model. However, this is kind of cheating since the Coulomb friction can be very different between components and can depend heavily on temperature, wear, and many more things. During our research on this, we found a paper on Model-free energy-based friction compensation [6], which described a way to estimate the friction while the robot arm is running. The main idea from this observer is that when the rotor kinetic energy and SEA spring potential energy return to zero at the end of a motion cycle, the net input $E_m = \sum P_{in} T_s$ over that cycle is exactly the same as the energy dissipated by the friction. The Coulomb and viscous coefficients are estimated by solving the 2×2 linear system over 2 consecutive cycles:

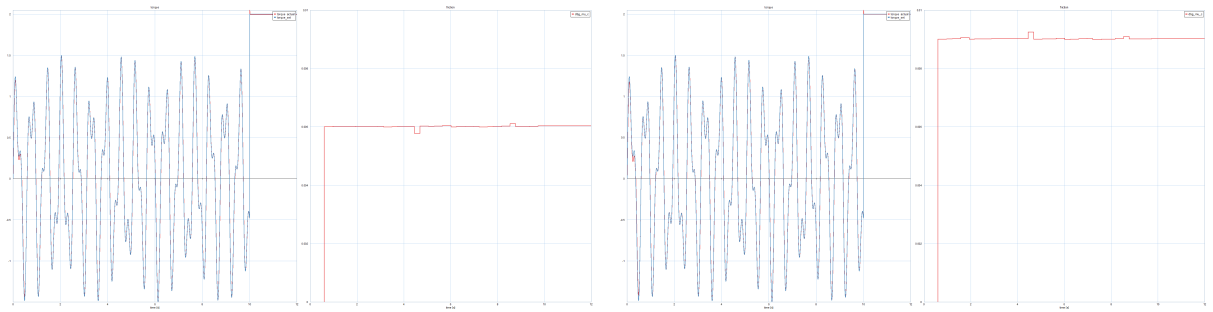
$$\begin{pmatrix} T_s \sum_{t=T_{cyc}^{n-1}}^{T_{cyc}^n} \text{sgn}(\omega_m) \omega_m & T_s \sum_{t=T_{cyc}^{n-1}}^{T_{cyc}^n} \omega_m^2 \\ T_s \sum_{t=T_{cyc}^n}^{T_{cyc}^{n+1}} \text{sgn}(\omega_m) \omega_m & T_s \sum_{t=T_{cyc}^n}^{T_{cyc}^{n+1}} \omega_m^2 \end{pmatrix} \begin{pmatrix} \hat{\mu}_c \\ \hat{\mu}_v \end{pmatrix} = \begin{pmatrix} E_{net}^n \\ E_{net}^{n+1} \end{pmatrix}$$

where E_{net}^n is the net cycle energy corrected for the previously applied compensation. The motion cycles are detected by the hysteresis on $|\omega_m|$, with initiation threshold $v_{init} = 0.2$ rad/s and termination threshold $v_{term} = 0.05$ rad/s.

To prevent instability from overcompensation, a Time Domain Passivity Algorithm (TDPA) monitors E_m and adds adaptive damping $\alpha = -E_m / (T_s \omega_m^2)$ whenever $E_m < 0$. The compensation output is then:

$$\hat{\tau}_{fric} = (\hat{\mu}_c \text{sign}(\omega_m) - \alpha \omega_m) \cdot N$$

In practice, the viscous estimate $\hat{\mu}_v$ was found to be very inaccurate and was therefore not implemented; the reason why this is so inaccurate is still unknown. We do speculate that it could be how we simulate the friction with the tanh function, but we have no proof for this. What, however, does work very well in this simulation is the Coulomb friction, and since this is the term that caused the constant error, we set out to implement this, the result of which can be seen in Figure 4. Here we gave the system a setpoint of $0.5 \cdot \sin(22.3) + \sin(10)$ to excite the system each cycle differently, and it goes over the 0 point multiple times. As can be seen from the simulation output in Figure 4, the friction gets estimated very accurately and changing the Coulomb friction by 0.003 also increased the estimate by 0.003, validating that if the Coulomb friction increases during a run (for example, by a change in temperature), the estimate compensates correctly.



((a)) Friction observer results at standard values ($0.000316 + 0.0057 = 0.006016$). ((b)) Friction observer results increased coulomb with 0.003 ($0.000316 + 0.0087 = 0.009016$).

Figure 4: Validation of Coulomb friction observer.

When we then implemented this into our 3-DoF robot arm, we, however, ran into some problems. When running the simulation, joint 1 worked perfectly as expected; however, this was not the case for joints 2 and 3. In the case of joint 2, it had a high peak right when an external load was added in our simulation, and in the case of joint 3, the Coulomb friction was never found during the run. Our theory on this is that the system must be excited well enough to calculate it accurately, and that sudden torque changes can cause wrong estimations. So in the case of joint 2 the overshoot is due to the high sudden torque output, and in the case of joint 3, it is never calculated because the force acting on it is always in the same direction due to our trajectory and gravity. Due to time restrictions, we weren't able to follow up on this further but our theory is that if the friction is made more robust by also checking the path it takes and the external load, the spikes can be reduced/mitigated. Furthermore, a startup sequence could be made that excites the system in a known way that can estimate the Coulomb friction at the startup and during a run when wanted to update a baseline, and then add limits during the run (for example, the Coulomb friction cannot be more than $\pm 20\%$ of the baseline).

2.3.3 Current Control

The current controller takes the measured motor speed and adds $K_e \omega_m$ straight into its output voltage command, on top of the PI feedback term. Where $K_e = K_t = 0.105 \text{ Nm/A}$ is the motor constant, since the back-EMF voltage is $V_{emf} = K_t \omega_m$. This feedforward correction cancels the disturbance before it has a chance to cause a tracking error instead of waiting for an error to appear and reacting to it. The PI controller then only needs to handle the difference in reference and measured current. The controller uses proportional gain $k_p = 1.335$ and integral time constant $\tau_I = 7.35 \times 10^{-4} \text{ s}$. This loop runs at 10 kHz, the faster of the three, because the motor's electrical time constant is by far the quickest dynamic anywhere in the system. The electrical time constant of the motor winding is $\tau_e = \frac{L_w}{R_w}$. With the values from Table 2, this gives $\tau_e = L_w/R_w = 5.71 \times 10^{-4}/0.762 \approx 0.75 \text{ ms}$, so the 10 kHz sample rate ($T_s = 0.1 \text{ ms}$) provides about 7 to 8 samples per electrical time constant, which is sufficient enough to track the current dynamics. Current can change almost instantly compared to how fast the link itself can move, which is exactly the kind of speed gap the cascaded control structure depends on.

PI control was chosen because the current loop sees a first-order electrical system ($V = Ri + L \frac{di}{dt}$). A first-order plant like this does not really have any oscillatory tendency at all, so there is nothing for a derivative term to damp. If we were to add one, it would amplify whatever noise is sitting on the current measurement without getting anything in return. The integral term, on the other hand, is what drives the steady-state current error to zero despite uncertainty in the exact values of R and L , and despite whatever the back-EMF feedforward does not perfectly cancel.

Both the current controller's output voltage and the current reference itself are saturated to realistic hardware limits: the current reference is clamped to the motor's rated continuous current of 3.84 A, and the output voltage is limited to the supply voltage of 48 V. Without these limits, the controller could command voltages or current the real driver could never produce.

2.3.4 Digital Discretisation

All three control loops are implemented in discrete time using a sample-and-hold pattern: at each sample instant the inputs are sampled, then the control law is computed, and finally, the output is held constant until the next sample. This is the standard Zero-Order Hold (ZOH) discretisation, which maps the continuous time integral $\int u dt$ to the forward Euler recurrence $u[k+1] = u[k] + T_s f[k]$. Forward Euler was chosen because it is explicit and therefore computationally cheap, which matters quite a bit at the 10 kHz current loop rate.

The three loops run at two distinct sample rates. The current loop runs at $f_s = 10 \text{ kHz}$ ($T_s = 0.1 \text{ ms}$), chosen to give about 7 to 8 samples per electrical time constant $\tau_e \approx 0.75 \text{ ms}$, which is the fastest dynamic in the system. The torque and impedance loops both run at $f_s = 1 \text{ kHz}$ ($T_s = 1 \text{ ms}$), which is a bit slower, but still fast enough to track the mechanical dynamics of the joints while leaving the current loop comfortably settled between updates.

The AD conversion block samples the joint positions and desired trajectory at 1 kHz and numerically differentiates the joint angles with a backward difference to obtain velocity. The DA conversion from

the current loop to the H-bridge introduces a half-sample computation delay of $T_s/2 = 0.05$ ms, which modelled explicitly in the simulation with a **DA-Delay** block.

2.4 Minimum Jerk Trajectory (MJT)

The minimum jerk trajectory is a model used that generates smooth, natural, human like movements by minimising the rate of change of acceleration (the jerk), which ensures that a robotic arm moves without any sudden, jerky adjustments [18]. To move the end-effector smoothly between the various Cartesian set points, the trajectory generator executes 4 sequential steps: inverse kinematics, motion timing, joint-space minimum jerk interpolation and a forward kinematics projection back into Cartesian space.

2.4.1 Inverse Kinematics

The set points are specified as Cartesian positions $\mathbf{x}_d = [x, y, z]^\top$. Before any interpolation can happen, these have to be converted into joint-space targets $\mathbf{q}_{\text{target}}$. For our yaw-pitch-pitch form, the inverse kinematics has a closed-form solution. The yaw angle comes directly from the horizontal target position: $q_1 = \text{atan2}(y, x)$. The 2 pitch joints form a planar two-link arm in the vertical plane at angle q_1 . The elbow angle can found by

$$\cos q_3 = \frac{r^2 + z'^2 - L_2^2 - L_3^2}{2 L_2 L_3}, \quad q_3 = \arccos(\cos q_3)$$

where $r = \sqrt{x^2 + y^2}$ is the horizontal reach, and $z' = z - (L_1 + z_{\text{base}})$ the height above joint 2. Since it is a positive root, the elbow-down configuration is selected. The shoulder angle is

$$q_2 = \beta - \alpha, \quad \beta = \text{atan2}(r, z'), \quad \alpha = \text{atan2}(L_3 \sin q_3, L_2 + L_3 \cos q_3)$$

To prevent the arm from moving outside the reachable workspace, $\arccos$ is clamped to $[-1, 1]$.

2.4.2 Motion Duration

The total motion time for each segment is set by dividing the largest joint displacement by average joint speed ($\bar{\omega} = 0.5$ rad/s):

$$t_f = \frac{\max_i |q_{\text{target},i} - q_{\text{start},i}|}{\bar{\omega}} + \epsilon$$

where $\epsilon = 0.001$ s makes sure division by zero does not happen when a set point coincides with the current position. This choice ensures that the slowest joint drives the timing and that no joint moves faster than $\bar{\omega}$.

2.4.3 Minimum Jerk Interpolation

The minimum jerk trajectory is performed in joint space. The MJT minimises the integral of the squared jerk ($\ddot{q}^2$) over the motion duration.

$$q_d(\tau) = q_{\text{start}} + (q_{\text{target}} - q_{\text{start}}) (6\tau^5 - 15\tau^4 + 10\tau^3)$$

The velocity profile is obtained by differentiating the above equation:

$$\dot{q}_d(\tau) = \frac{q_{\text{target}} - q_{\text{start}}}{t_f} (30\tau^4 - 60\tau^3 + 30\tau^2)$$

To make sure no velocity or acceleration step is happening at the start or end of a segment. This MJT is applied to all 3 joints independently and simultaneously with a the shared motion duration.

2.4.4 Forward Kinematics

Since the impedance controller expects a Cartesian reference position and velocity. These are obtained by applying forward kinematics to the MJT:

$$\begin{aligned}x_d &= \cos q_{d,1} (L_2 \sin q_{d,2} + L_3 \sin(q_{d,2} + q_{d,3})) , \\y_d &= \sin q_{d,1} (L_2 \sin q_{d,2} + L_3 \sin(q_{d,2} + q_{d,3})) , \\z_d &= L_1 + L_2 \cos q_{d,2} + L_3 \cos(q_{d,2} + q_{d,3})\end{aligned}$$

The Cartesian velocity is then gotten by differentiating these equations.

3. Simulation

3.1 Verification

Before integrating the full controller, the individual subsystems were isolated and checked against expected physical behaviour. This follows the principle that a positive cause should produce the correct sign of effect, and that a passive system should dissipate rather than gain energy.

3.1.1 Not actuated arm under gravity

The first check verifies the rigid-body dynamics and the gravity direction of the manipulator. To verify this subsystem, the manipulator was separated from the actuators: the SEAs were removed and each joint was driven with zero torque, so the arm is completely unactuated and only gravity acts on it. The arm is released from a horizontal pose ($q_2(0) = 1.5708$ rad).

Because gravity acts purely along $-z$, it exerts no moment about the vertical yaw axis, so the yaw joint q_1 should not move. The two pitch joints should swing down from the release pose. The isolated arm has no dissipative elements, since the bearing friction and SEA flexure damping all live in the actuators that were removed, so the manipulator forms a conservative system. It should therefore oscillate about its hanging equilibrium without losing amplitude, rather than settling. A non-decaying oscillation is in fact the stronger check here: a growing amplitude would mean the model spuriously injects energy, while a decaying one would point to hidden numerical damping. Constant-amplitude motion confirms that the rigid-body dynamics conserve energy as they should.

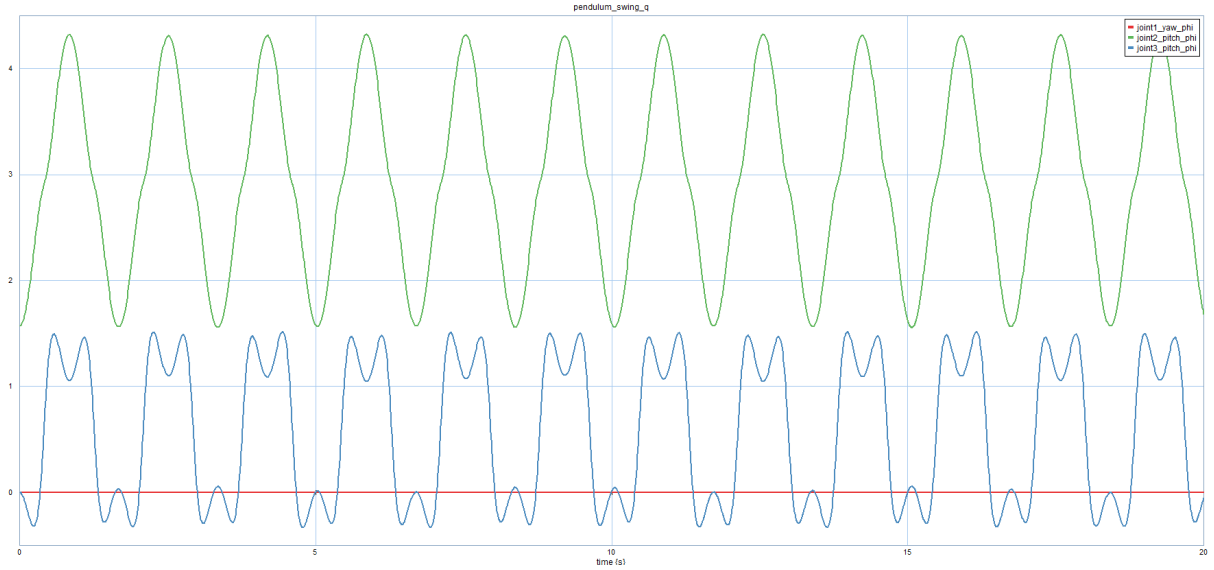


Figure 5: Unactuated arm released from a horizontal pose.
Generated with `G01_pendulum_test.emx`.

The simulation in Figure 5 confirms this. The yaw joint q_1 stays at 0 rad throughout, confirming that gravity is applied purely in the $-z$ direction and is decoupled from the yaw axis. The two pitch joints oscillate continuously about the hanging equilibrium without their amplitude decaying, which confirms that the manipulator conserves energy and that the gravity direction and rigid-body behaviour are correct before any actuation is added.

3.1.2 Gravity compensation

The second check verifies that the gravity feed-forward term holds the arm against its own weight. The arm is commanded to a single, statically gravity-loaded pose, a forward horizontal reach at $\mathbf{x}_d = [0.35, 0, 0.55]^T$ m, and held there with the nominal Cartesian impedance gains ($K = 400$, $D = 65$). The

experiment is run twice: once with the gravity feed-forward enabled ($g_{\text{on}} = 1$) and once with it disabled ($g_{\text{on}} = 0$).

With compensation enabled, the feed-forward term supplies the joint torques needed to balance gravity, so the impedance controller only has to reject small residual errors and the end-effector should reach the commanded height. With compensation disabled, the impedance stiffness alone must carry the full gravity load, so the arm settles below the target at a steady-state offset of approximately $\Delta z \approx F_g/K$.

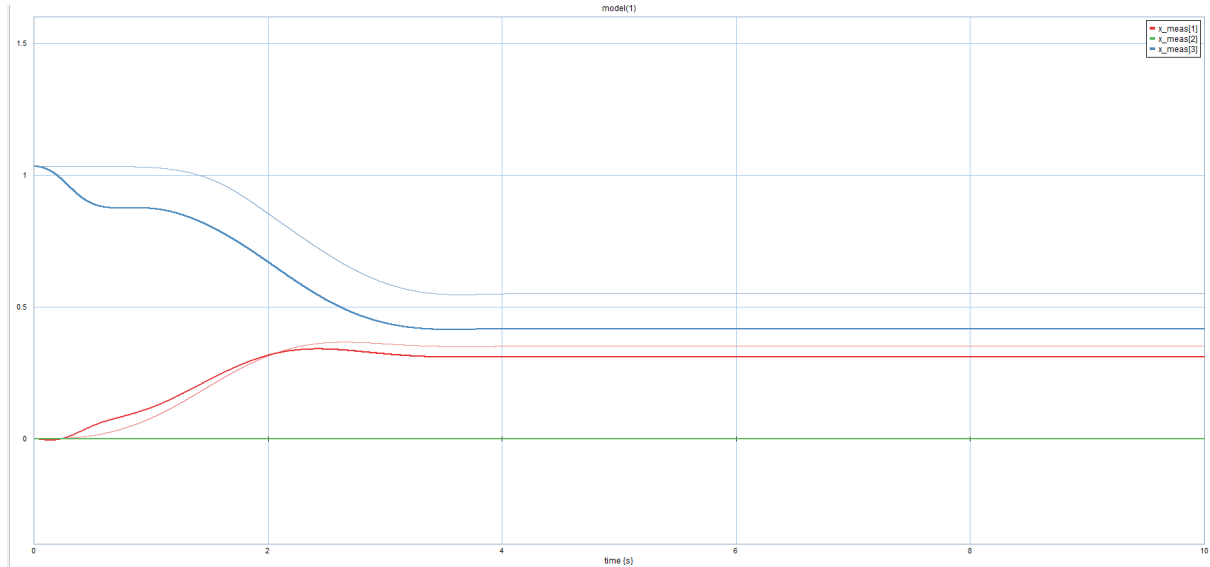


Figure 6: End-effector height while holding a fixed gravity-loaded pose, with and without gravity compensation. Generated with `G01_finalmodel.emx`.

Figure 6 confirms the expected behaviour. With compensation enabled the end-effector settles at 0.549 m, a steady-state error of only 1 mm. With compensation disabled the end-effector droops to 0.416 m, a steady-state offset of 134 mm below the target. The difference between the two runs demonstrates that the gravity feed-forward correctly cancels the gravitational load on the manipulator, which is what allows the impedance gains to be kept low for safe, compliant interaction rather than being raised to fight gravity.

3.1.3 Actuator loops: current, torque and speed

The third check verifies the two inner control loops of a single actuator and inspects whether the resulting currents, torques and speeds are realistic. The internal signals of joint 1 (the yaw joint) are recorded while the arm performs the reaching task. Several test runs with different settings were carried out to observe the actuator behaviour; the run shown in Figure 7 is representative. External contact forces are disabled for this experiment so that the actuator behaviour is observed on its own.

During the smooth parts of the motion both loops track their references closely. The torque loop uses the SEA spring deflection as its torque sensor, comparing the commanded joint torque τ_{ref} against the measured spring torque $K_{\text{SEA}} \Delta\theta$; the measured torque follows the reference with a typical magnitude of about 0.7 N m. The current loop, running an order of magnitude faster, keeps the measured current on its reference, staying below about 0.2 A away from the transitions. The motor speed, measured on the motor side of the gearbox (before the 1:120 reduction), peaks at roughly 115 rad s^{-1} . This corresponds to an output speed at the joint of about 0.96 rad s^{-1} , well within the 2.62 rad s^{-1} joint speed limit.

At the set point transitions, short spikes appear in the joint torque (up to about 3.5 N m) and in the motor current (up to about 5.5 A). These spikes are present in the torque *reference* itself and not only in the measured signal. This shows that they originate upstream of the actuator, at the moments where a new set point is applied, rather than being a fault of the inner current or torque loop. The inner loops in fact reproduce these reference transients faithfully. We were not able to conclusively determine the

exact mechanism that produces the spikes; they are brief and coincide consistently with the set point transitions. The current spikes momentarily exceed the 3.84 A continuous rating of the motor, but only as short transients at the transitions, while the current stays far below the rating during the rest of the motion. This behaviour is revisited in the discussion.

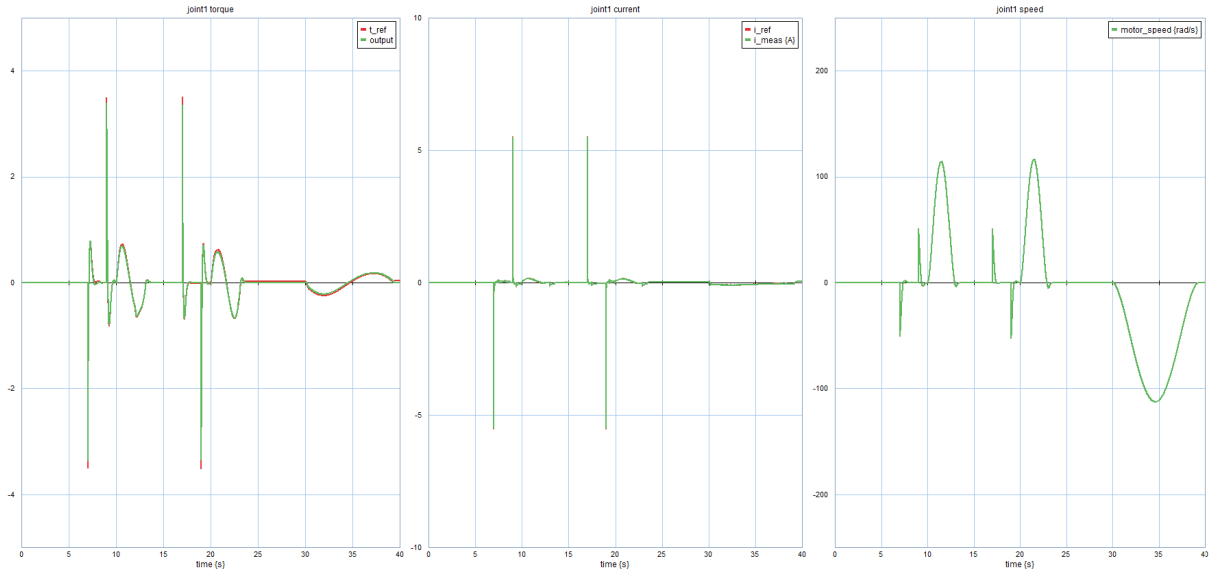


Figure 7: Joint 1 actuator verification during the reaching task. Left: commanded vs measured joint torque. Middle: current reference vs measured current. Right: motor speed. Generated with `G01_finalmodel.emx`.

3.2 Validation

After verifying the individual subsystems, the full model is tested on the reaching task it was designed to perform. The arm starts in its home configuration and is commanded to visit four Cartesian set points in sequence ($\mathbf{p}_1 = [0.35, 0, 0.55]$, $\mathbf{p}_2 = [0, 0.5, 0.25]$, $\mathbf{p}_3 = [0, 0.5, 0.55]$, $\mathbf{p}_4 = [0, -0.35, 0.45]$, all in metres). The trajectory between set points is shaped by the minimum-jerk generator in joint-space and projected back to Cartesian space. The controller is active with nominal gains ($K = 400$, $D = 65$, $g_{\text{on}} = c_{\text{on}} = 1$) and no external contact force is applied.

3.2.1 Decoupled response to external forces

The first validation experiment checks that the Cartesian impedance controller behaves as a diagonal spring in the world frame: a force applied along one axis should produce a steady-state deflection along that same axis only, with a magnitude given by $\Delta x = F_{\text{ext}}/K$, and combined forces should superpose linearly. The arm holds the commanded set point with nominal gains ($K = 400$, $D = 65$, $g_{\text{on}} = c_{\text{on}} = 1$) while an external force of 10 N is applied to the end-effector in four successive intervals: first along $+x$ only, then along $+y$ only, then along $+z$ only, and finally along all three axes simultaneously.

The left panel of Figure 8 shows the measured end-effector position against the commanded set point and the right panel shows the component-wise tracking error $\mathbf{e}_x = \mathbf{x} - \mathbf{x}_d$. During each single-axis interval the end-effector deflects by approximately 25 mm on the loaded axis, while the other two components stay at their commanded values within a few millimetres. This matches the predicted deflection $\Delta x = F_{\text{ext}}/K = 10/400 = 0.025$ m to the millimetre and confirms that the impedance gain acts as a pure spring with the prescribed stiffness. When all three forces are applied at the same time in the last interval, a 25 mm deflection appears on each axis simultaneously, which shows that the responses superpose linearly without cross-coupling between the Cartesian directions.

The small residual error of a few millimetres visible on the unloaded axes is consistent with the residual gravity-compensation error already observed in Section 3.1.2, where a 1 mm steady-state offset remained even without external load. Together these observations confirm that the Cartesian impedance law, the

Jacobian-transpose mapping and the gravity feed-forward correctly implement a diagonal spring at the end-effector in world coordinates, which is the property that makes the impedance controller predictable and tunable axis by axis.

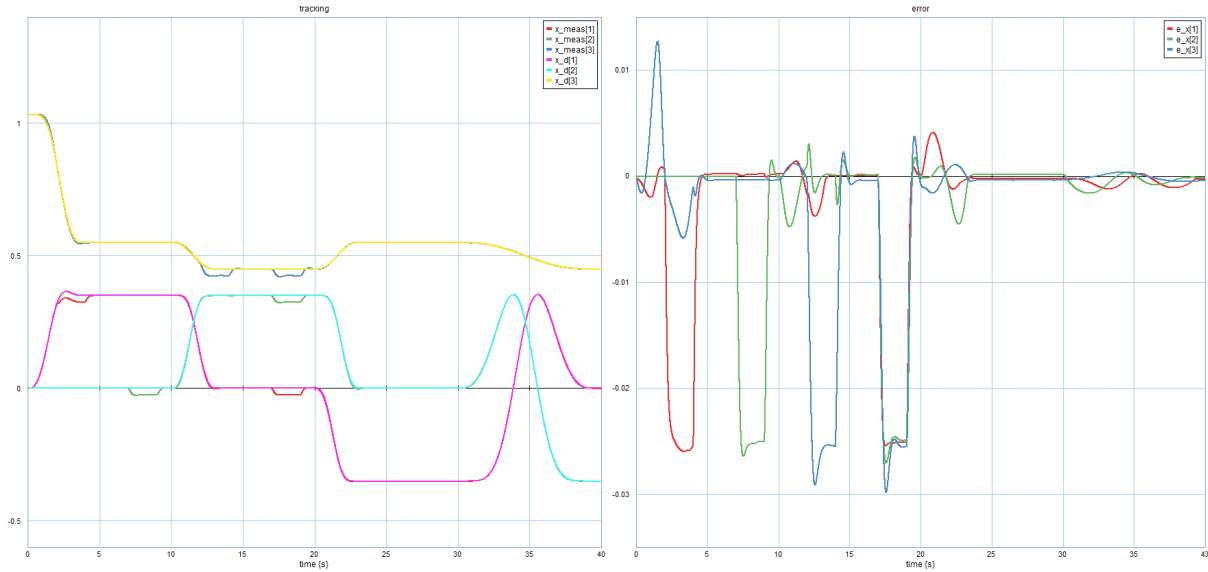


Figure 8: End-effector response under axis-aligned external forces. A 10 N force is applied to the end-effector in four successive intervals: $+x$ only, $+y$ only, $+z$ only, and finally all three axes simultaneously. Left: measured end-effector position $\mathbf{x}$ and commanded set point $\mathbf{x}_d$ for each Cartesian component. Right: component-wise tracking error $\mathbf{e}_x$.

Generated with `G01_finalmodel.emx`.

3.2.2 Interaction with an external load

The second validation experiment demonstrates the compliant behaviour of the Cartesian impedance controller under an unmodelled external load. While the arm performs the reaching task, a payload of $m_{\text{block}} = 2 \text{ kg}$ is attached to the end-effector between $t = 19 \text{ s}$ and $t = 30 \text{ s}$, applying a constant downward force of $F_z = -mg \approx -19.6 \text{ N}$ in the world frame. The payload is intentionally not included in the gravity feed-forward, so the impedance law alone must absorb its weight. All other settings are nominal ($K = 400$, $D = 65$, $g_{\text{on}} = c_{\text{on}} = 1$).

Figure 9 shows the end-effector tracking (left) and the joint 2 torque reference (right). The payload is applied near the end of the $\mathbf{p}_2$ hold and is then carried through the transition to $\mathbf{p}_3$, which is held throughout the rest of the loaded interval. During the loaded hold on $\mathbf{p}_3$ the end-effector settles at a vertical position of 0.40 m while the commanded height is 0.45 m , a steady-state droop of 50 mm . This is the expected behaviour of a spring-damper at the end-effector: the impedance reaches a new equilibrium where the virtual restoring force balances the extra weight. After the payload is removed at $t = 30 \text{ s}$ the end-effector recovers cleanly back up to the commanded height. Because $\mathbf{p}_4$ has the same vertical coordinate as $\mathbf{p}_3$, the recovery is observed in isolation, without the load release being mixed in with a simultaneous change in the commanded height. There is no significant deviation in the x and y components, confirming that the disturbance acts purely in the vertical direction as imposed.

The joint 2 torque on the right panel of Figure 9 shows the underlying actuation effort. During the unloaded hold on $\mathbf{p}_2$ the torque settles at about -10.2 N m , balancing the arm's own weight through the gravity feed-forward. During the loaded hold on $\mathbf{p}_3$ the torque settles at about -16.4 N m , an increase of roughly 6.2 N m . This matches the static moment that a 2 kg payload produces about the shoulder pitch axis at this configuration ($mg \cdot r \approx 6 \text{ N m}$ for a moment arm of about 0.3 m), which independently verifies that the actuator is delivering the physically expected joint torque. The peak torque remains far below the joint's 87 N m continuous rating, confirming that the actuator operates well within its capabilities while the arm complies safely with the contact load.

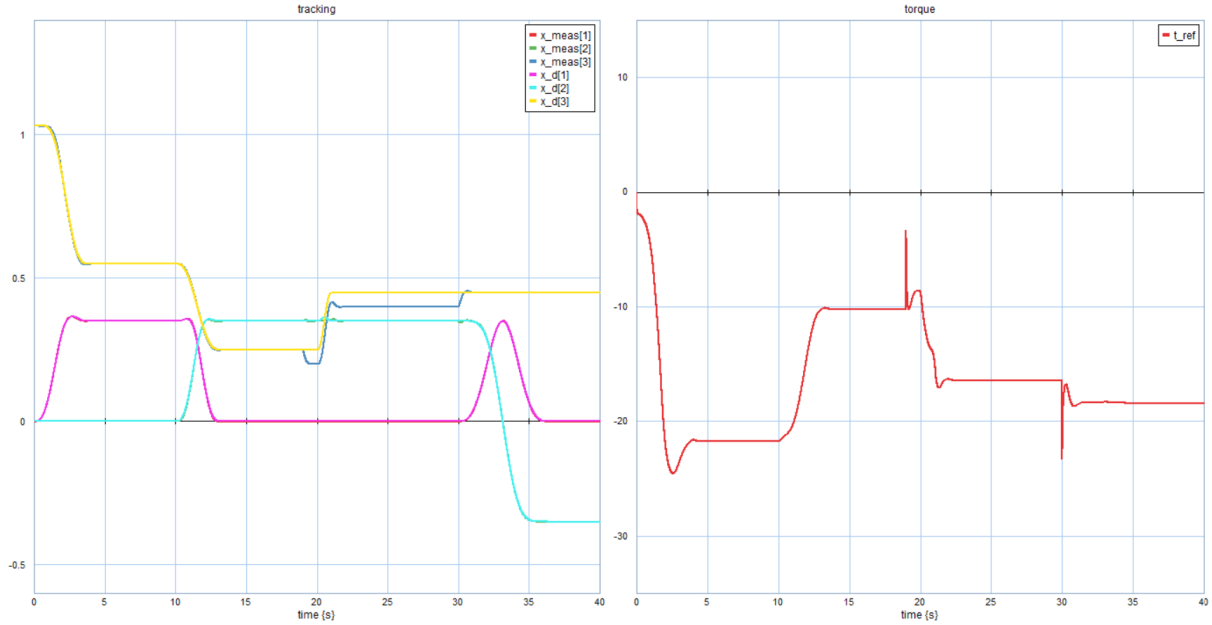


Figure 9: Reaching task with a 2 kg external payload applied between $t = 19$ s and $t = 30$ s. Left: end-effector position $\mathbf{x}$ and commanded set point $\mathbf{x}_d$. Right: joint 2 torque reference.

Generated with `G01_finalmodel.emx`.

3.2.3 Stiffness variation under the same external load

The previous experiment used a single Cartesian stiffness of $K = 400$. To demonstrate the design trade-off introduced by the impedance gain, the same 2 kg payload is now applied while the arm performs the same reaching task at three different stiffness settings: $K = 150$, $K = 400$ and $K = 800$. The damping is held at $D = 65$ in all three runs.

The three end-effector height responses are shown in Figures 10, 11 and 12. During the loaded hold on $\mathbf{p}_3$ the soft setting $K = 150$ produces a steady-state droop of approximately 130 mm below the commanded height, the nominal setting $K = 400$ gives a droop of about 50 mm, and the stiff setting $K = 800$ reduces the droop further to roughly 25 mm. As expected, a stiffer impedance gives way less under the same external force.

The numerical pattern is the more interesting result. The product of the stiffness and the droop is approximately constant across the three runs: $150 \cdot 0.130 \approx 19.5$ N, $400 \cdot 0.050 \approx 20.0$ N and $800 \cdot 0.025 \approx 20.0$ N. All three values closely match the payload weight $mg = 19.6$ N, which is the force the impedance spring must balance in steady state. This shows that the end-effector indeed behaves as a linear Cartesian spring at this configuration, and that the relation $\Delta z = F_g/K$ holds across more than a factor of five in stiffness. This is a quantitative validation of both the impedance law and the Jacobian-based mapping that turns the virtual Cartesian force into joint torques.

The experiment also makes the design trade-off explicit. A soft impedance yields safe, compliant behaviour, since an unexpected contact deflects the arm by several centimetres before producing large reaction forces, but it also accepts a sizeable steady-state error under any sustained load. A stiff impedance tracks the target much more accurately under load but pushes back harder on contact, losing some of the safety benefit that impedance control is meant to provide. The nominal $K = 400$ sits between these extremes, trading a moderate 50 mm droop under a 2 kg payload for the ability to give way safely on contact.

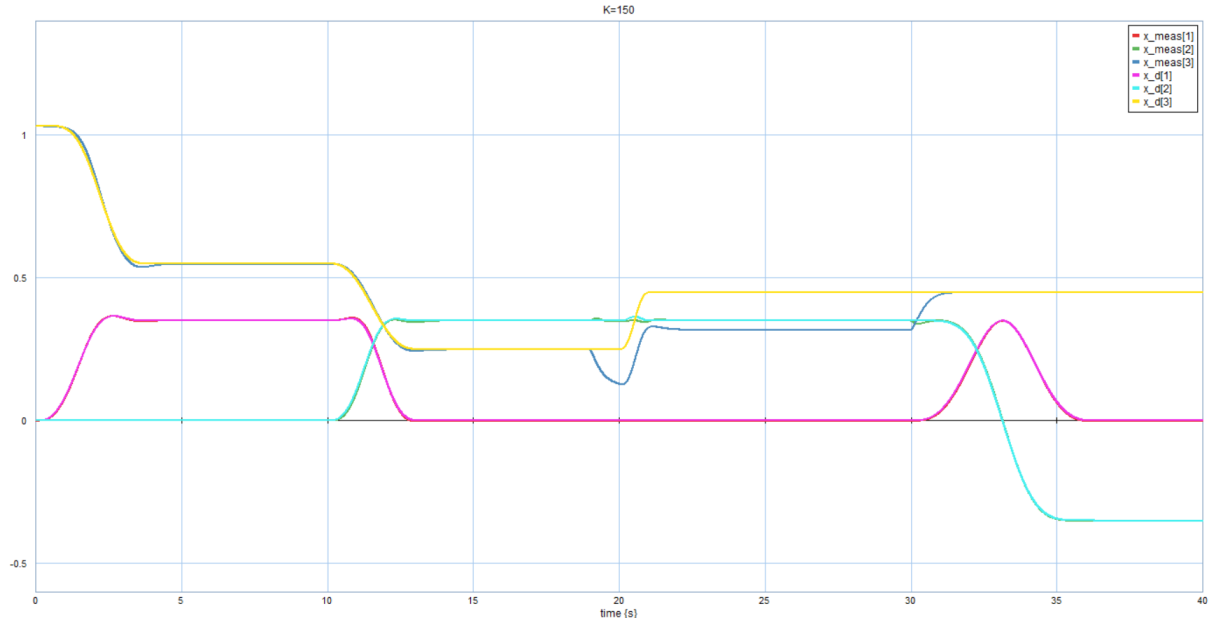


Figure 10: End-effector tracking with the 2 kg payload at the soft setting $K = 150$.
Generated with *G01_finalmodel.emx*.

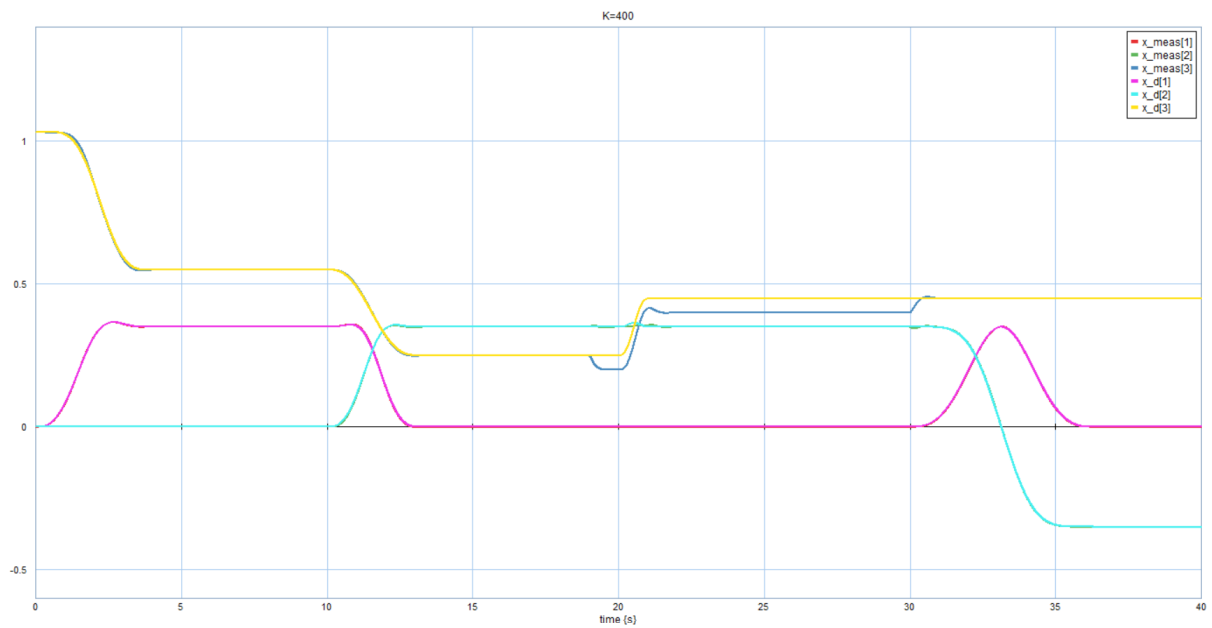


Figure 11: End-effector tracking with the 2 kg payload at the nominal setting $K = 400$.
Generated with *G01_finalmodel.emx*.

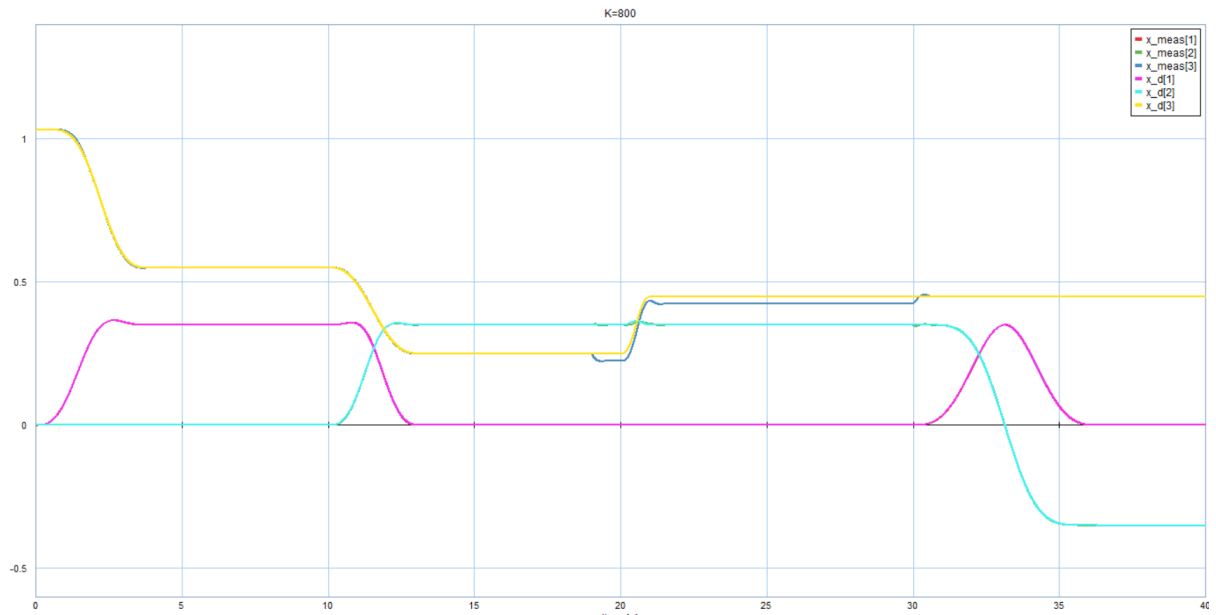


Figure 12: End-effector tracking with the 2 kg payload at the stiff setting $K = 800$.
Generated with *G01-finalmodel.emx*.

4. Discussion & Conclusion

Discussion

Overall, we managed to include several realistic physical effects into our 20-sim model quite successfully. By adding the motor and gearbox characteristics and non-linear friction curves, our model does a great job of simulating how the real 3 DOF robotic arm behaves.

Part of the success of this implementation is the integration of the gravity compensation. By deriving a closed-form Coriolis matrix and mapping the gravity vectors directly through link-specific Jacobians. That way the Cartesian impedance controller is completely relieved from fighting the arm's own weight.

This allowed us to keep the robotic arm safe and human-centred like the original FR3. The validation experiments confirmed that the cascaded loop structure functions effectively; when interacting with an unexpected external load, the end-effector responds as a true spring-damper, smoothly giving way on contact rather than unsafely resisting at full power.

However, looking back critically, there are a few shortcomings and simplifications that did not completely go according to plan:

- **Reference torque spikes:** As mentioned in 3.1.3, during setpoint transitions, we noticed some spikes in the reference torque ($\tau_{\text{ref}} \approx 3.5 \text{ N} \cdot \text{m}$), which temporarily pushed the motor current up to around 5.5 A on Joint 1. This behaviour actually comes directly from the command signal itself, not from our inner control loops. Even though this brief current surge goes over the continuous rating of 3.84 A, it is still safely within the motor's transient pulse-rating limits. We could not pinpoint the exact cause for sure, but we think the trajectory supervisor is accepting new setpoints mid-motion, which creates acceleration discontinuities in the Minimum Jerk Trajectory.
- **Off-diagonal inertia terms:** Because of the limits within the 3D Mechanics Editor, we had to leave out the off-diagonal inertia terms from our model. This means we only looked at the main rotation axes. These missing terms represent the cross-coupling forces, they show how moving one joint creates a pulling force or twist on another joint. Because the weight of the robot parts is mostly balanced and centred along the main axes, these cross forces are very small. We think leaving them out will be a negligible change to the simulation results.

Conclusion

The project showed us that bond-graph modelling is a great way of analysing and designing mechatronic systems. It was a real challenge to bridge the gap between the theoretical mechanics, hardware limitations and design choices. We were able to make a realistic simulation model that closely represents its real-world behaviour.

In terms of what we learned, we found ourselves understanding the core principles of modelling more and more as we progressed through the assignments. Moving from simpler models to this project gave us a clear insight into the differences between a 2-DOF and a 3-DOF robotic arm, especially regarding how much more complex the multi-axis dynamics and control become. Furthermore, we gained an appreciation and understanding for the design of human-centred robotics. Working with Series Elastic Actuator elements taught us how compliant joints can make robots significantly safer and more adaptable when interacting with humans.

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Appendices

A Motor Efficiency Analysis

To evaluate the electro-mechanical conversion losses within the DC motor, an efficiency analysis was performed. A simulation was done under a dynamic load profile to map the internal winding resistance against the speed-dependent friction of the rotor. The bondgraph utilized is illustrated in Fig. 13.

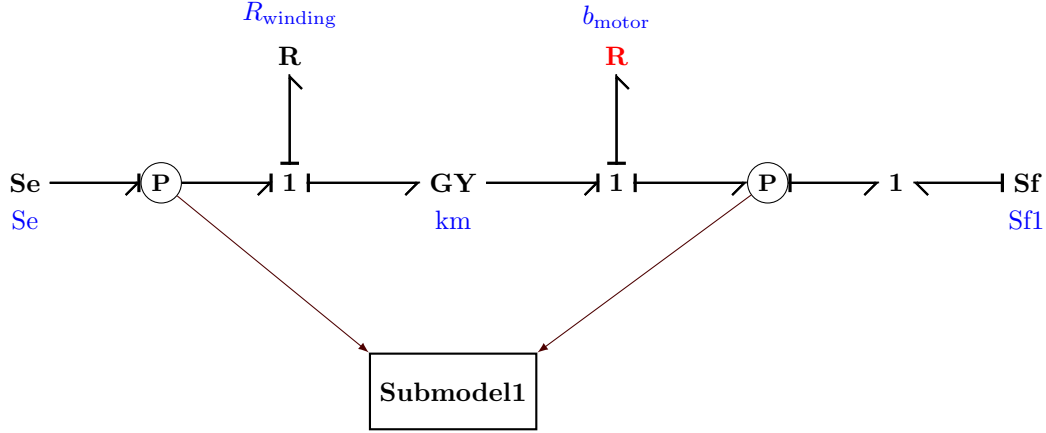


Figure 13: Bondgraph model for the motor efficiency test

Simulation Model Configuration

The total voltage drop and internal energy dissipation are split into electrical losses via the armature winding resistance (R_{winding}) and rotational losses via the non-linear rotor friction component (b_{motor}). The armature resistance follows a linear relation:

$$U_R = r_{\text{winding}} \cdot I \quad (2)$$

The mechanical torque loss ($\tau_{f,\text{motor}}$) acting on the internal 1-junction includes both viscous damping and an empirical Coulomb approximation implemented via a hyperbolic tangent smoothing function:

$$\tau_{f,\text{motor}} = b_{\text{motor}} \cdot \omega + T_{c,\text{motor}} \cdot \tanh(\gamma \cdot \omega) \quad (3)$$

where ω represents the angular velocity of the rotor shaft. The transformation between electrical and mechanical power domains is dictated by the motor constant (K_m) acting as a gyrator (GY). The physical parameters are summarized in Table 3.

Table 3: DC Motor Submodel Simulation Parameters

Symbol	Parameter Description	Value	Unit
r_{winding}	Armature Winding Resistance	0.762	Ω
$r (K_m)$	Motor Torque Constant (Gyrator)	0.105	$\text{N} \cdot \text{m}/\text{A}$
b_{motor}	Rotor Viscous Friction Coefficient	1.27×10^{-5}	$\text{N} \cdot \text{m} \cdot \text{s}/\text{rad}$
$T_{c,\text{motor}}$	Rotor Coulomb Friction Torque	0.0057	$\text{N} \cdot \text{m}$
γ	Hyperbolic Tangent Shape Factor	5000	—

Analysis of Efficiency Curves

A simulation was executed under a constant voltage source ($Se = 48 \text{ V}$), while the motor output shaft was driven by a linearly increasing flow source (Sf_1) to replicate dynamic load variation over time. The non-linear behavior reveals operational zones:

- **Low-Speed Friction Dominance:** At the start of the acceleration profile ($t < 2 \text{ s}$), the efficiency curve goes steeply from 0%. In this zone, the static Coulomb threshold ($T_{c,\text{motor}}$) dominates the losses.

- **Peak Efficiency Window:** The motor achieves its maximum efficiency of approximately 92% around $t = 9.7$ s. At this point, the power conversion through the gyrator is optimally balanced against the internal Joule heating (I^2R) losses of the copper windings.
- **High-Load Armature Saturation:** As the flow demand scales towards the 10-second mark, the efficiency undergoes a steep, non-linear degradation down to roughly 78. This rapid decay highlights high-load current saturation, where resistive dissipation inside R_{winding} dominates over mechanical output generation.

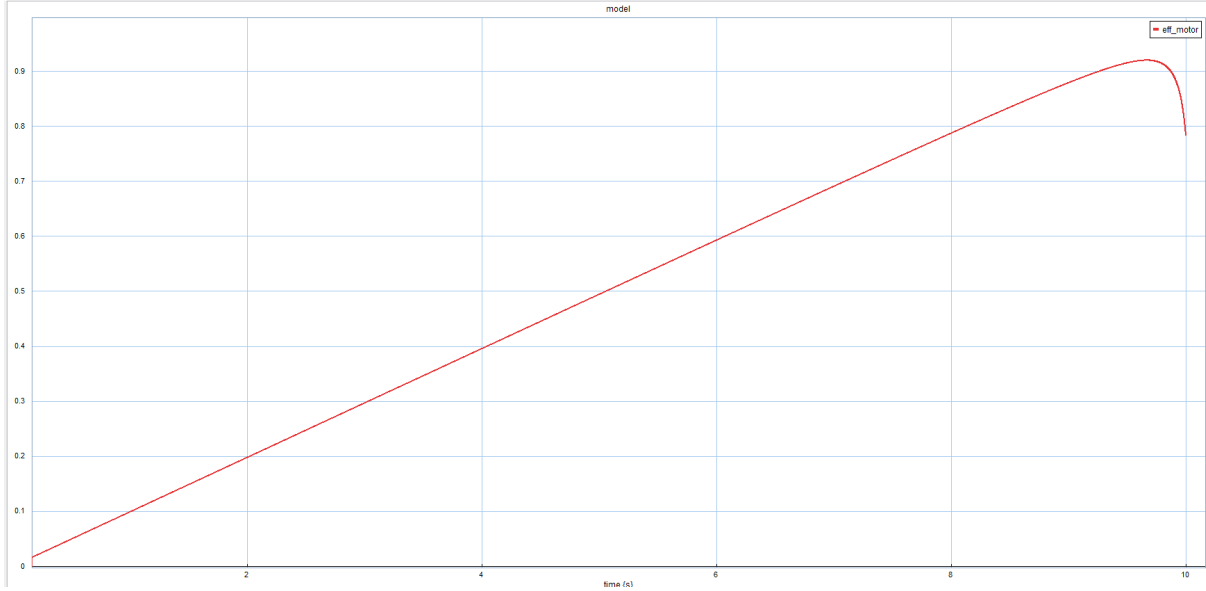


Figure 14: simulation response showing the non-linear motor efficiency (η_{motor}) tracking over a 10-second progressive loading ramp

B Gearbox Efficiency Analysis

To validate the transmission and losses, an efficiency analysis was performed on the simulated drivetrain. By isolating the gearbox transmission, the relation between the joint output speed, power dissipation, and the overall efficiency was evaluated. The bondgraph, which can be seen in Figure 15.

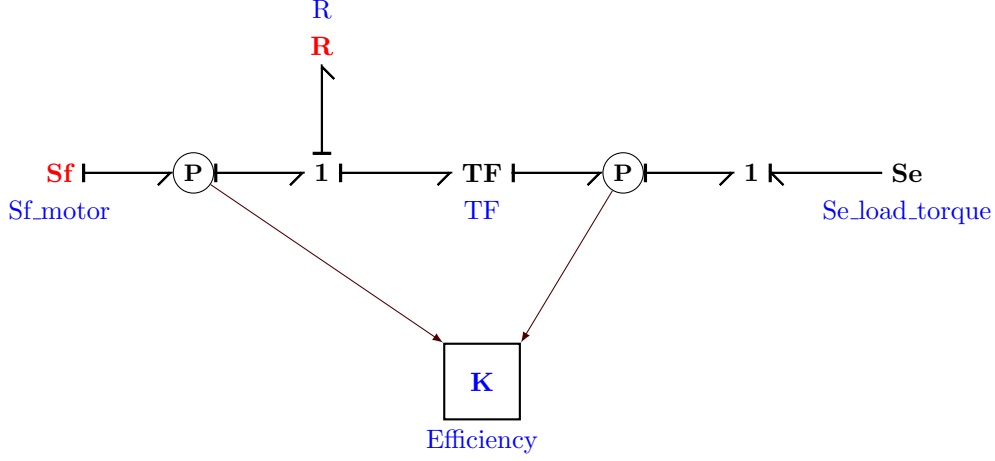


Figure 15: bond graph model configuration for the transmission efficiency test

Simulation Model Configuration

The internal loss characteristics of the gearbox submodel consist of constant friction and speed-dependent damping. The total friction torque (τ_f) within the gearbox is implemented using the following relation:

$$\tau_f = d_v \cdot \omega + d_c \cdot \tanh(5000 \cdot \omega) \quad (4)$$

where ω represents the angular velocity of the joint. The constant Coulomb friction coefficient (d_c) was taken from the datasheet [13], whereas the viscous friction coefficient (d_v) was calibrated dynamically. The parameters utilized for this experiment are summarized in Table 4.

Table 4: 20-sim Gearbox Submodel Simulation Parameters

Symbol	Parameter Description	Value	Unit
N	Gearbox Ratio	120	—
d_c	Gearbox Coulomb Friction	0.038	$\text{N} \cdot \text{m}$
d_v	Gearbox Viscous Friction	1.75×10^{-4}	$\text{N} \cdot \text{m} \cdot \text{s}/\text{rad}$

Analysis of Efficiency Curves

A simulation was executed, during which the motor input velocity was linearly accelerated. The resulting power balances and non-linear efficiency curves demonstrate two insights regarding drivetrain behavior:

- **Speed-Dependent Loss Inefficiency:** Under ideal zero-speed starting conditions, the gearbox achieves a peak efficiency of approximately 92%, where the friction losses are predominantly restricted to the Coulomb friction (d_c). However, as the output velocity goes towards the robot's operating range, the speed-dependent viscous friction (d_v) increasingly dissipates energy.
- **Power Divergence:** The input power (p_{in}) required by the drivetrain diverges non-linearly from the actual utilized output power (p_{out}) delivered to the load. This growing power gap confirms that high-ratio strain wave gearboxes experience pronounced energy dissipation in the form of structural heat during high-speed trajectory tracking.

By including these realistic loss curves, this ensures that when designing joint torque controllers later on, they are tested against realistic motor limits and power drops. The results of this can be seen in Figure 16.

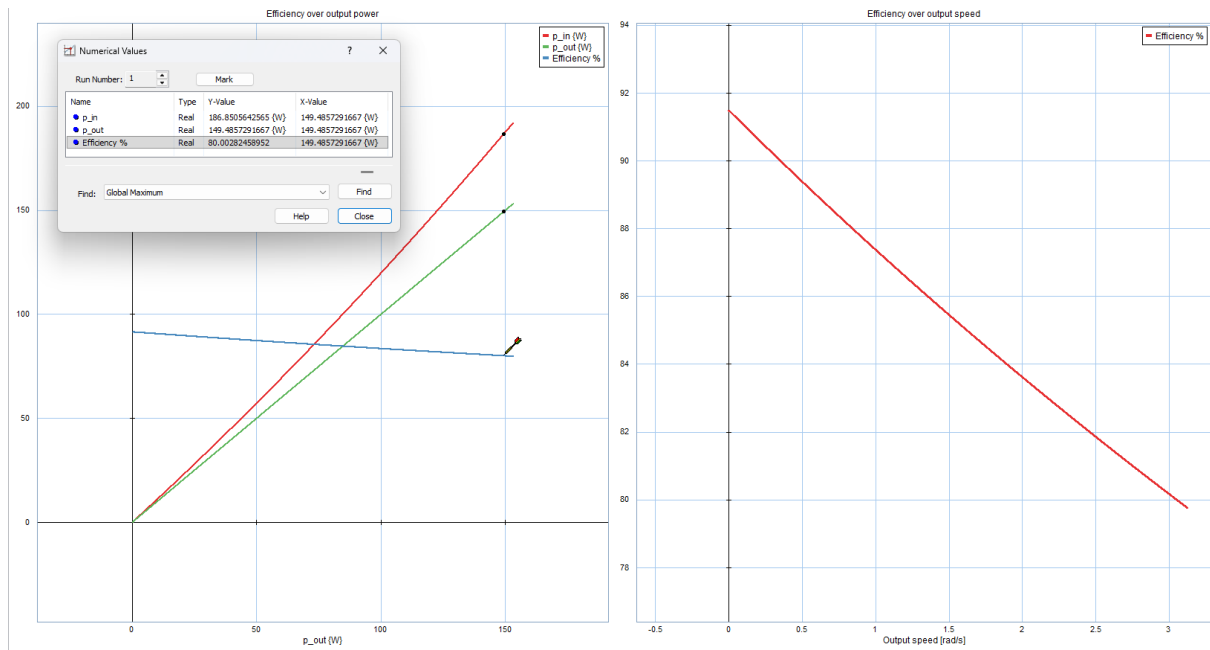


Figure 16: Simulation results showing output power vs. input power (left) and the linear efficiency degradation over joint output speed (right)

C Inertia properties franca robot FR3

Table 5: Parameters of the 3-DoF (yaw–pitch–pitch) reduced model of the Franka Research 3 as used in the 20-sim model. Reduced-model joints 1–3 correspond to FR3 joints 1, 2 and 4; FR3 joints 3 and 5 are locked at 0 rad and joints 6 and 7 are removed (mass set to zero). Body 2 merges FR3 links 2–3, body 3 merges links 4–5. All values are taken from the official `franka_description` package. The frame offset is the translation of the joint frame with respect to the previous joint frame (body 1: with respect to the base); centre of mass and inertia tensor are expressed in the link’s own local frame.

Parameter	Unit	Joint/body 1 (yaw)	Joint/body 2 (pitch)	Joint/body 3 (pitch)
<i>Kinematics</i>				
Rotation axis	–	Z	Y	Y
Frame offset o_x	m	0	0	0.0825
Frame offset o_y	m	0	0	–0.316
Frame offset o_z	m	0.333	0	0
<i>Rigid body</i>				
Mass m	kg	2.9275	5.1804	4.9427
Centre of mass c_x	m	0.0000	0.0194	–0.0639
Centre of mass c_y	m	–0.0181	–0.1665	0.1683
Centre of mass c_z	m	–0.0386	0.0029	–0.0183
Inertia I_{xx}	kg m ²	0.023927	0.123830	0.148269
Inertia I_{yy}	kg m ²	0.022482	0.046217	0.047631
Inertia I_{zz}	kg m ²	0.006350	0.140787	0.152528
Inertia I_{xy}^a	kg m ²	0.000013	0.013211	0.020178
Inertia I_{xz}^a	kg m ²	–0.000114	0.007133	0.014891
Inertia I_{yz}^a	kg m ²	–0.001995	–0.005810	–0.005939
<i>Actuation and joint dynamics</i>				
Gear ratio n (reduction)	–	120	120	120
Rotor inertia J_m	kg m ²	4.206399×10^{-5}	4.206399×10^{-5}	3.211626×10^{-5}
Joint damping b	N m s/rad	0.003	0.003	0.003
Coulomb friction τ_c (URDF)	N m	0.2	0.2	0.2
Viscous friction μ_v^b	–	16.0	16.0	16.0
Coulomb friction μ_c^b	–	0.0	0.0	0.0

^a The 20-sim 3D Mechanics Editor supports diagonal inertia tensors only; the off-diagonal terms are listed for completeness and are omitted in the implemented model (documented modelling assumption).

^b Friction coefficients taken from the FR3 dynamics parameter file; units are not specified in the source data and should be verified before these values are used in the simulation model.